

Organ Disease Progression Visualizer: An Imaging-Driven 3D Lung Disease Progression Model

Final Report
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Important Note. The table below reflects the contribution (as %) made by different team members to the work carried out in this project the preparation of this document. The name entered in the last column will legally count as a digital signature for internal purpose at MQ. By putting your signature here you are honourably declaring the quantum of the contribution you have made to the project/report in the relevant aspect(s).

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ABSTRACT:

Pulmonary fibrosis progressively destroys lung tissue, and yet traditional diagnostic tools rarely connect flat imaging data to intuitive 3D visuals. This project bridges that gap by introducing a deterministic, CT-driven pipeline that transforms raw chest CT scans into interactive, anatomically accurate 3D lung models. The system extracts three quantitative disease signals directly from patient data tissue density severity, structural texture, and high-density ratio and maps them directly to precise transformations on a 3D lung mesh. Because every visual shift is strictly constrained by real imaging evidence, there is no guesswork involved, and a structured audit trail logs every change to ensure complete clinical accountability. Validated across eight distinct patient cases, this interactive system reliably demonstrated reproducible, explainable progression from early to advanced disease states with auditable logs that explained, with grounded evidence, the rationale behind the visual depictions. The end result was a prototype tool that illustrated clear and meaningful links between complex medical imaging data and spatial anatomy.

1. Introduction

1.1 Aims

The primary objective of this research was to develop a deterministic framework that bridges the gap between qualitative clinical narratives and quantitative 3D spatial data. We aimed to establish a pipeline that utilises Large Language Models (LLMs) to interpret pathological markers within clinical reports and map these findings directly to coordinate transformations in a 3D pulmonary mesh. This research focused on creating an auditable visual progression of Idiopathic Pulmonary Fibrosis (IPF), ensuring every anatomical modification is traceable to specific clinical evidence, a concept we have defined as "clinical accountability".

1.2 Background

Pulmonary fibrosis is a chronic and progressive lung disease where healthy lung tissue is gradually replaced by stiff scar tissue. As this condition advances, the lungs inevitably lose their natural elasticity and overall volume decreases. As such, the body's ability to transfer oxygen into the bloodstream starts to severely deteriorate. In its most common form, Idiopathic Pulmonary Fibrosis (IPF), the underlying cause remains completely unknown. Therefore, the condition is both irreversible and ultimately fatal, making it absolutely critical to understand how the disease distributes spatially across the tissue and changes over time to improve diagnosis, patient education, and clinical planning.

In respiratory internal medicine, the gold standard for diagnosis remains High-Resolution Computed Tomography (HRCT), an advanced version of the standard CT scan, which provides tens or even hundreds of 2D imaging slices of lung tissue that traverse the organ along the transaxial plane. Translating these slices into a holistic 3D understanding of disease progression, however, is cognitively demanding for clinicians and patients alike (Vasilescu et al., 2021). Recent advancements in 3D medical modelling, such as the characterisation of fibroblast foci, have demonstrated that 3D volumetric analysis offers superior insights into the spatial distribution of lesions compared to traditional 2D imaging (Jones et al., 2016)

To address this limitation, recent medical research has pushed for a shift toward three-dimensional analysis. For instance, Jones et al. (2016) used micro-CT imaging to create

3D characterisations of fibroblast foci, the foundational clusters of scar tissue in IPF. Their work proved that 3D volumetric analysis offers vastly superior prognostic information compared to traditional 2D measurements. Furthermore, Jacob et al. (2019) argued that for spatially distributed conditions like pulmonary fibrosis, 2D slice-based imaging provides an inherently fragmented view that simply cannot capture the true shape and extent of the disease.

Simultaneously, the rise of domain-specific LLMs, such as ClinicalBERT, has revolutionised the extraction of structured data from unstructured clinical notes (Huang et al., 2019). While these models excel at identifying disease phenotypes (Tustison et al., 2024), a significant technical gap remains: most AI-driven visualisations are generative, whereby the images are created from scratch with no grounded reference point, which leads to medical inaccuracies i.e. hallucinations, that lack clinical veracity (Dinc, M. T., et al, 2025). Our research sits at this intersection, moving beyond simple text classification (Lu et al., 2024) toward the spatial manipulation of anatomical meshes.

In response to this need, this project introduces a deterministic, CT-driven pipeline that maps quantitative diagnostic signals directly onto a 3D lung mesh. Instead of forcing doctors to mentally piece slices together, our system looks to automatically extract three vital disease metrics from raw chest CT scans: tissue density severity, structural texture, and the high-density ratio. These numerical signals are then translated into precise, vertex-level deformations and surface changes on an anatomically accurate reference model. By leveraging advanced geometric libraries alongside reference meshes, the pipeline dynamically morphs the 3D model to reflect true pathology. Crucially, every visual change is strictly constrained by empirical imaging data, and an unalterable audit trail logs every modification to ensure complete clinical accountability. Ultimately, when validated across eight real patient cases, this interactive system demonstrates a highly reproducible, explainable progression from healthy tissue to advanced fibrosis, bridging the gap between raw data and spatial anatomy.

1.3 Research Problems and the Context

Despite these recent advances, a significant and largely unaddressed gap still exists at the intersection of medical imaging and 3D visualisation. Crucially, the problem is not an absence of tools; rather, it is the complete absence of a principled, transparent connection between raw imaging data and the final visual output.

Currently, most AI-driven medical visualisation systems are generative in nature, that is, these models learn statistical patterns from large datasets and produce outputs that look highly plausible, however, the relationship between any specific input scan and the resulting visual remains entirely opaque. For instance, Bhadra et al. (2021) demonstrated that generative approaches, such as Generative Adversarial Networks and diffusion models, frequently produce medical images that are visually convincing but clinically inaccurate. In these systems, lesions can appear in anatomically implausible locations, and disease patterns often fail to correspond to the actual input data. Furthermore, there is no built-in mechanism for a clinician or educator to verify why the output looks the way it does. Consequently, Wang et al. (2024) identified this interpretability gap as a fundamental barrier to deploying AI

in high-stakes medical contexts, arguing that healthcare systems require absolute transparency and human-centred design before these tools can be considered truly reliable.

A second and closely related problem centers on determinism. In longitudinal clinical monitoring, where tracking a patient's disease over time is the priority, it is absolutely essential that the same input always produces the exact same output. If identical CT data can yield visually different representations depending on the unpredictable, stochastic behaviour of a generative model, the tool becomes useless for any purpose requiring consistency, whether that is clinical, educational, or research-oriented.

Therefore, this project aims to directly identify and address both of these critical limitations. The core challenge encapsulating these is what is often termed: the semantic-to-spatial mapping problem. Often, CT imaging produces highly quantitative but spatially abstract data, while 3D mesh deformation requires precise mathematical parameters defined at the level of individual mesh vertices. Ultimately, bridging these two distinct representations in a principled, reproducible, and auditable way, so that every single visual change can be traced directly back to a specific imaging measurement, is the central technical contribution of this work.

1.4 Expected Outcomes

To address these distinct limitations, this project sets out to achieve four core outcomes (outlined below), in which each were fully realised and are discussed in indepth detail across Sections 1.2 and 1.3.

The primary deliverable was a functional, three stage 3D progression model that demonstrated a continuous and smooth transition from healthy lung morphology, through early and moderate stages, to advanced fibrosis. Crucially, this model is driven entirely by real CT imaging data rather than manually defined variables. Therefore, the visual representation reflects authentic patient geometry at every phase.

Building on this model, the second outcome establishes a quantitative imaging to 3D mapping method. This method serves as a formal specification of how CT extracted signals directly control the visual and structural parameters of the lung mesh. For instance, it details the weighting formula used to compute a composite progression score, alongside the mathematical functions that translate that score into lesion spread, tissue colouration, surface roughness, and localised structural deformation.

Furthermore, our third objective introduces a trust first transformation pipeline where every mesh modification is not only justified by empirical imaging evidence but also recorded in a structured, machine readable audit trail. This audit trail constitutes the primary clinical accountability mechanism of the system, creating a permanent, per case record that explicitly links each visual alteration back to the specific CT signal that caused it.

Finally, the fourth outcome comprises a portfolio grade interactive demonstration designed to present these breakthroughs effectively and intuitively. This interface features a dynamic disease progression slider, a live clinical reasoning panel that updates in real time as the slider moves, and comprehensive, case by case explanations of dominant disease drivers and active visual effects. Ultimately, we designed this tool in a way that we envision it being useful, and

highly suitable, for demonstration to both academic and professional audiences as well as the general public at large.

1.5 Benefits and Significance

The broader significance of this project lies in its clear demonstration that medical visualisation does not need to choose between visual quality and interpretability. Crucially, the system that we have built shows that a relatively small set of quantitative CT features, in our case derived from a cohort of eight real patient cases, is completely sufficient to drive a 3D model that produces visually differentiated, stage-appropriate, and clinically grounded representations. Because these features are extracted using established image processing techniques, the pipeline requires no deep learning model and no manual annotation whatsoever.

For medical education, a tool of this kind offers a much more intuitive way to communicate the spatial reality of diseases such as pulmonary fibrosis than traditional CT slices alone. Often, as is the case with pulmonary fibrosis, looking at individual slices makes it difficult to perceive the overall impact on the organ. For patients, therefore, it provides a means of understanding what is physically happening to their lungs in a way that is both accessible to a non-specialist and entirely honest and grounded in underlying data. For researchers and developers, the project demonstrates a highly viable architecture for explainable medical visualisations. Consequently, this framework could easily be extended to other organs, other diseases and other, richer, imaging inputs such as full volumetric CT stacks.

Perhaps, most importantly, the audit logging system developed in this project provides a concrete implementation of the clinical accountability principle. This is the core idea that in medical software, every output must be explainable, every transformation must be traceable, and every decision must be defensible. This principle is increasingly recognised as essential for the future deployment of computational tools in healthcare. Ultimately, this project offers a working prototype of what that accountability looks like in practice.

2. Literature Review

The body of literature selected for this review was chosen as it directly related to the three core technical domains involved in our project: 3D volumetric lung simulation, biomedical natural language processing, and interpretability in medical machine learning. To maintain a narrow focus on the problems we were trying to address, research and studies pertaining to general-purpose generative art architectures, non-medical mesh animation pipelines, and purely statistical 2D survival models were consciously excluded. These fields do not account for the structural constraints or clinical verification frameworks required when handling authentic human anatomical data. Instead, the selected studies were chosen because, combined, they map a precise path whereby one could go from a qualitative clinical text entry to a validated, auditable spatial modification.

2.1 The Shift from 2D Imaging to 3D Volumetric Modelling

The clinical necessity for moving beyond 2D imaging techniques is well-documented in current academic literature. Traditional clinical frameworks in respiratory internal medicine

rely heavily on multi-slice High-Resolution Computed Tomography (HRCT) to identify and track interstitial lung diseases. However, converting a massive stack of these 2D spatial cross-sections into a comprehensive assessment of longitudinal tissue decay presents a severe cognitive bottleneck for medical practitioners. Weatherley et al. (2019) establish that advanced quantitative and experimental imaging techniques must look beyond the flat boundaries of traditional 2D slice analysis. Because pathologies like idiopathic pulmonary fibrosis (IPF) exhibit highly asymmetric, volumetric behavior, slice-by-slice evaluation frequently obscures the physical continuity of tissue texturing changes across distinct lung regions. Research by Jones et al. (2016) leveraged high-resolution micro-CT imaging to map the complete 3D micro-spatial distribution of fibroblast foci, demonstrating that explicit 3D volumetric modeling yields a significantly more accurate predictor of patient survival rates than traditional 2D surface area measurements. Furthermore, Bompoti et al. (2021) demonstrated that high-resolution volumetric imaging of lung tissue allows micro-CT frameworks to extract distinct morphology patterns, proving that structural macro-deformations and direct spatial orientation dictate how a clinician interprets organ degradation. Consequently, for any modern disease visualisation tool to remain clinically useful to a medical professional, it cannot rely on flat overlays or isolated 2D classifications; instead, it needs to operate entirely within a flexible 3D mesh environment capable of localised volumetric manipulations.

2.2 Extracting Structured Features from Clinical Texts

In addition to 3D visualisations, a key focus area of our research pertains to extracting structural findings from qualitative clinical notes. In a clinical setting, unstructured, narrative driven, clinical descriptions serve as the primary method for relaying direct visual image observations into actionable diagnostic interpretations. Tavakoli and Kim (2025) note that generating and compiling precise clinical histories directly within the radiology workflow closes historic communication gaps, which improves longitudinal tracking across separate hospital units. For a 3D visualisation engine, however, these reports cannot simply be summarised into other natural language texts; instead, they must be broken down into explicit spatial variables. Previous studies in natural language processing show that clinical text can be reliably converted to structured variables. Lybarger et al. (2021) demonstrated that domain-specific, i.e. medical, architectures can extract highly detailed radiological findings alongside normalised anatomical information using span-based relation extraction models. Their work confirms that raw text strings like "ground-glass opacities in the lower right lobe" can be decomposed into a standardised structural format: a density indicator (opacity), a specific pathological marker (ground-glass), and a bounded anatomical constraint (lower lobe). This systematic parsing provides the exact numerical parameter inputs required to guide localised downstream geometry modifications.

2.3 Large Language Models & Semantic to Spatial Gaps

The use of Large Language Models has been the subject of evolving research in the medical field. The baseline for capturing deep bidirectional context within unstructured text sequences began with standard encoder frameworks like BERT, introduced by Devlin et al. (2019). While general-purpose transformers manage basic syntactic dependencies efficiently, they lack the domain-specific exposure required to navigate the dense, abbreviated jargon found in hospital reporting systems. To close this operational gap, Huang, Altosaar and Ranganath (2019) developed ClinicalBERT by training representations directly on the extensive,

deidentified MIMIC-III clinical note database. Their findings proved that language models exposed to authentic medical terminology outperform general architectures by wide margins when mapping complex latent relationships in healthcare records. Furthermore, Tustison, Avants and Stone (2024) demonstrated that biomedical large language model latent spaces can successfully project a patient's clinical trajectory along a mathematical model, or "pseudotime," effectively mapping early states to advanced mortality markers. Similarly, Lu and Naseem (2024) explored leveraging disease network link predictions within massive language architectures to forecast complex progression pathways. Despite this promising research, however, a significant technical disconnect emerges when attempting to map these parsed semantic extractions directly onto physical coordinate adjustments. Neither model provides an integrated structural translation mechanism to project these higher-order semantic vectors onto actual 3D Cartesian coordinates. Trinh et al. (2026) evaluated how medical multimodal large language models process spatial datasets. They confirmed that, while these systems can categorise conditions textually, they consistently struggle to ground their interpretations within exact 3D coordinate spaces.

2.4 Generative Hallucinations vs. Clinical Determinism

As outlined in our project context, a debate exists as to whether generative or deterministic visualisations are more appropriate in the healthcare setting. High-stakes clinical modeling requires a total shift away from stochastic pixel generation toward traceable geometric manipulation. Image reconstruction research by Bhadra et al. (2021) analysed the mechanics of hallucinations within tomographic reconstructions, demonstrating that deep neural architectures that prioritise pure visual plausibility over empirical data frequently invent geometric artifacts completely unsupported by raw imaging evidence. As documented by Huang, L., et al. (2025) in a comprehensive taxonomy of model errors, unconstrained generative networks are highly prone to hallucinating plausible, but structurally fabricated, visual features. Dinc et al. (2025) further surveyed large language models across active healthcare settings, emphasising that medical inaccuracies and ungrounded statements present substantial risks to clinical safety. If an algorithm would generate highly realistic tissue textures but position fibrotic scarring in healthy, unaffected segments, as an example, it fails the basic requirement of medical accountability. This risk emphasises the need for strict interpretability guidelines, as established by Wang et al. (2024), who argue that complex clinical visualisations must maintain a transparent, auditable trail linking output structures directly back to input parameters.

2.5 Methodological Implications for System Design

Some modern approaches have attempted to guide geometric deformations using language constraints. Sun et al. (2025) introduced a framework for handle-based mesh deformation guided directly by vision-language models, showcasing that coordinate transformations can be influenced by contextual text inputs. Their pipeline, however, relies on iterative optimisation loops that do not guarantee strict determinism or clinical accountability. If the exact same text report yields slightly different coordinate translations across separate runs, the tool becomes completely unreliable for tracking exact longitudinal changes. Consequently, text processing layers need to function exclusively as structured parameter extractors that can pass bounded coordinates and intensity values to rule-based, human reviewed, geometric transformation engines. This research directly informed our project's choice of PyVista and Trimesh over unconstrained rendering suites like Blender, reflecting

this absolute need for clinical transparency. While Blender offers superior visual realism, its complex rendering pipelines are difficult to automate programmatically and fail to provide reproducible vertex-level data logs. Shifting to PyVista and Trimesh gave our team complete control over coordinate shifts, normal vector calculations, and boundary constraints. By driving deformations through rule-based functions that applied directly to an anatomically validated base mesh from the Human Atlas API, we believed we were able to eliminate generative hallucinations. In our progression model, every contraction, surface indentation, and localised density shift is a deterministic function of empirical clinical data. This methodology ensures complete auditability and addressed, in part, the literature's call for visual accountability in high-stakes medical informatics.

3. Research Plan and Methodology

3.1 Analysis - Decomposing the Problem

The central challenge of this project was translating medical imaging evidence into an interpretable, spatially meaningful 3D representation of disease progression. Rather than attempting to solve this as a singularly large problem, our team elected to decompose it into five discrete sub problems, each of which could be developed, tested, and validated independently before integration.

The first sub problem focused on CT image processing. This stage addressed how to reliably isolate lung regions from chest CT images of varying quality, format, and acquisition parameters. The system uses a consistent and reproducible algorithmic approach that requires no manual intervention whatsoever.

Building on this, the second sub problem involved feature extraction. Having successfully isolated the lung region, the challenge was how to extract a small set of quantitative, interpretable signals. These metrics meaningfully represent the degree and character of the disease present in the image, making them suitable for use as inputs to a downstream visualisation model.

The third sub problem centered on 3D model representation. This component determined how to obtain or construct an anatomically accurate 3D lung model to serve as a stable geometric substrate for disease simulation. Consequently, the team focused on how to load and manipulate that model programmatically in a way that fully supports the required transformations.

Following the model preparation, the fourth sub problem tackled disease simulation. This area explored how to translate the extracted imaging signals into specific, geometrically coherent visual and structural changes in the 3D model. Often, in simulations, these modifications must be proportional to the severity of the detected disease, spatially concentrated in the anatomically appropriate regions, and visually differentiated across the range of disease stages from healthy to advanced.

Finally, the fifth sub problem addressed interaction and visualisation. This step considered how to present the 3D model and its associated disease information to the user. Ultimately, the interface must be interactive, self explanatory, and highly suitable for demonstration to both technical and non technical audiences.

This strategic decomposition proved essential to managing the technical risk and maintaining forward progress throughout the project. Because each sub problem had clear inputs and outputs, progress on one component could continue independently of any delays or complications in another. As discussed in Section 3.5, this modular structure also made it possible to substitute alternative approaches when an initial strategy proved unworkable, without disrupting the overall pipeline.

3.2 Synthesis - Integrating the System

The overall system architecture follows a linear pipeline structure in which each stage produces outputs that become the inputs to the next ,as outlined in Figure 1:

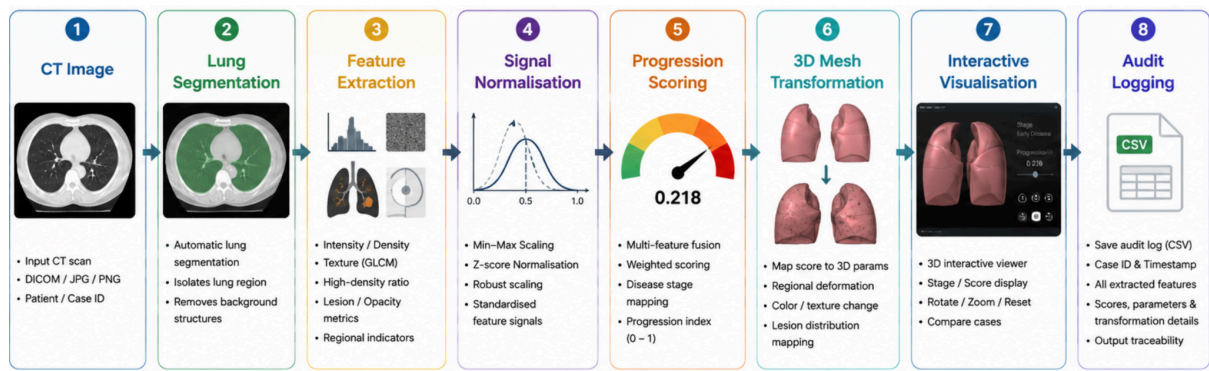


Fig1: Overall Integrated Lung Disease Progression Pipe Line

The CT image processing and feature extraction stages are implemented in a dedicated function, `extract_ct_features()`, which accepts any JPEG or PNG chest CT image as input and returns three quantitative signals: severity, texture, and high density ratio. Consequently, these signals are passed to `map_features_to_parameters()`, which normalises them against known clinical ranges and computes a weighted progression score along with four derived visualisation parameters: colour strength, roughness strength, lesion spread, and deformation strength.

These parameters are then used to configure three core simulation functions. For instance, `compute_disease()` generates a per vertex disease field on the lung mesh using Gaussian based spatial functions weighted toward the subpleural and basal regions. Following this, `deform()` applies four geometrically distinct transformations to the mesh vertices in proportion to the local disease field value. Furthermore, `make_tissue_colors()` maps the disease field to a smooth colour gradient from healthy pink through early red to advanced dark fibrotic tissue.

The interactive viewer, built using PyVista, presents the transformed mesh alongside a live explanation panel that updates in real time as the user moves the disease progression slider. To maintain accountability, the `ProgressionAuditLogger` class records the full pipeline trail,

including CT extraction, feature mapping, stage transitions, and vertex modification counts, to a structured JSON file that is saved when the session ends.

The integration of these components was completed incrementally, as described in Section 3.3, with each stage validated before being connected to the next. Therefore, the final integrated pipeline is invoked with a single command:

```
python "Integrated Pipeline.py" case_01_image.jpg --save_audit
```

Ultimately, this single command runs the complete pipeline from raw CT image to interactive 3D visualisation, and produces a permanent audit log of every decision made along the way.

3.3 Methodology - Development Process

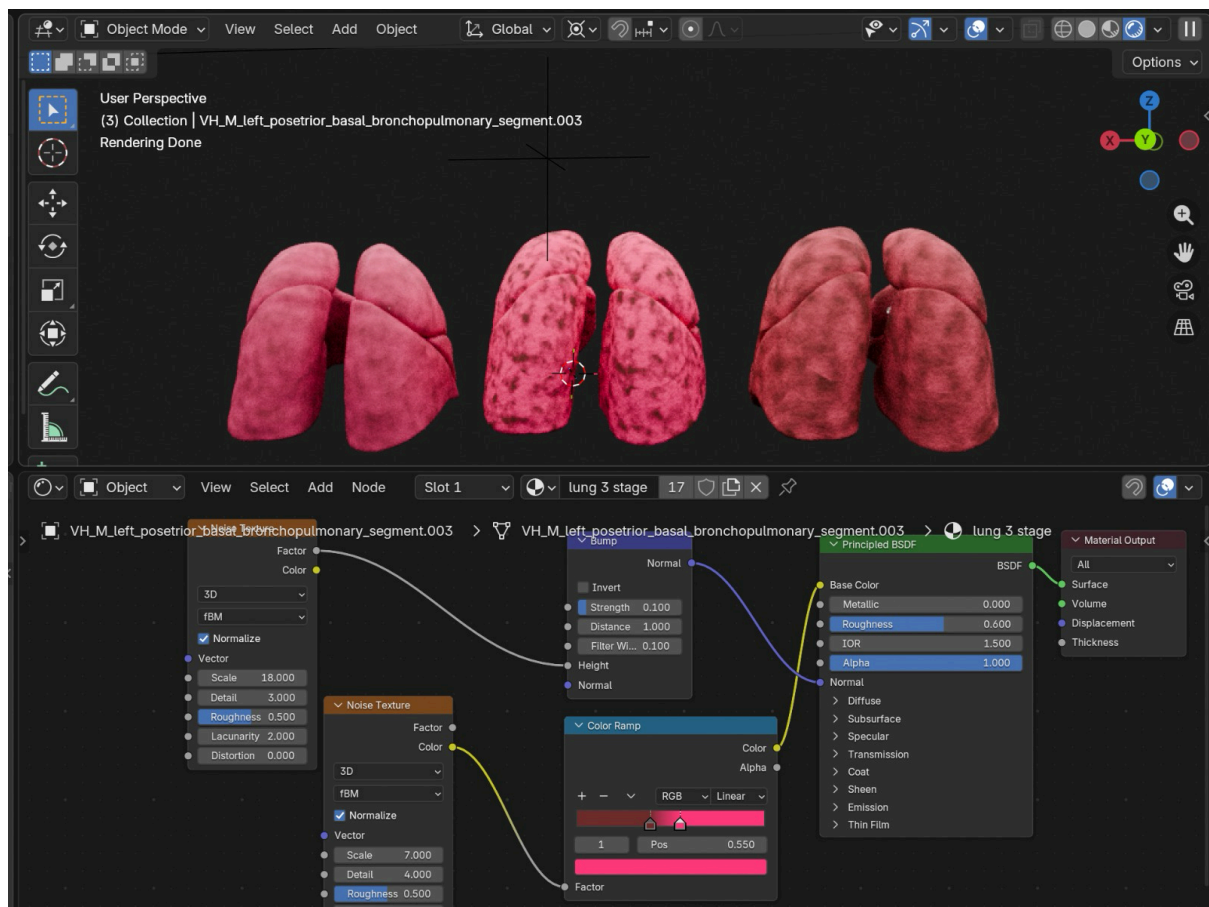
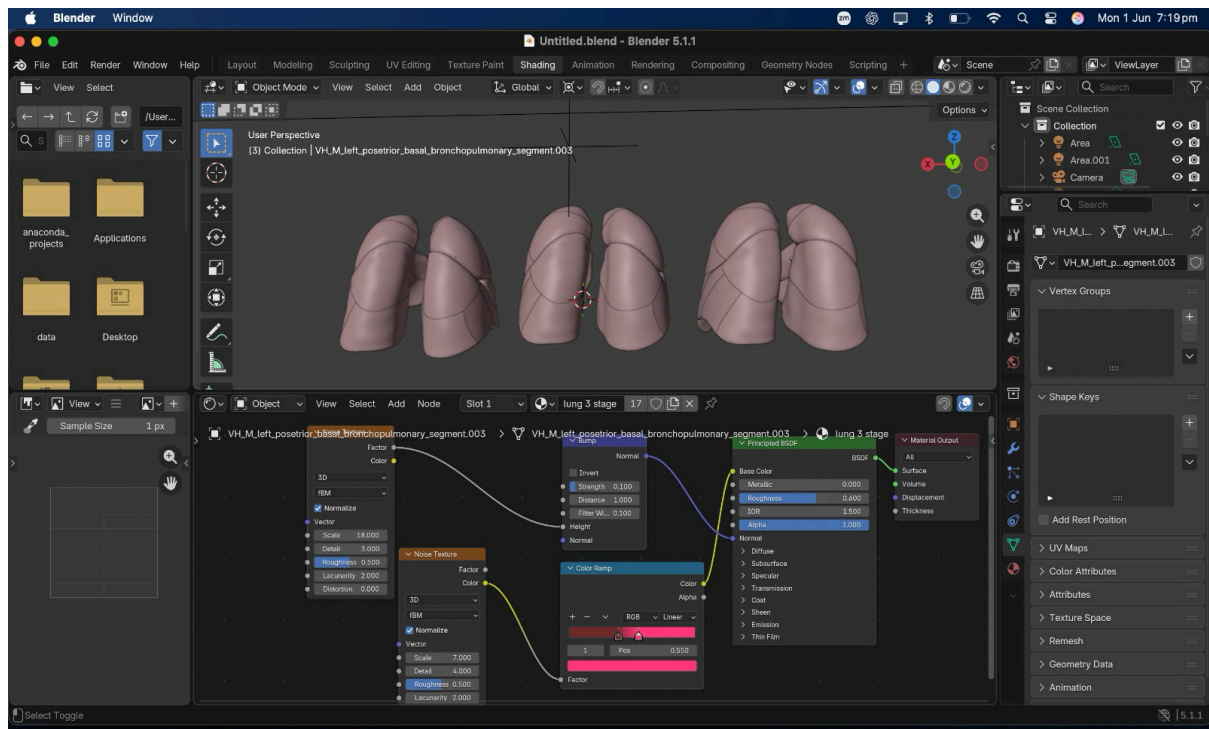
3.3.1 Initial Exploration and the Blender Attempt

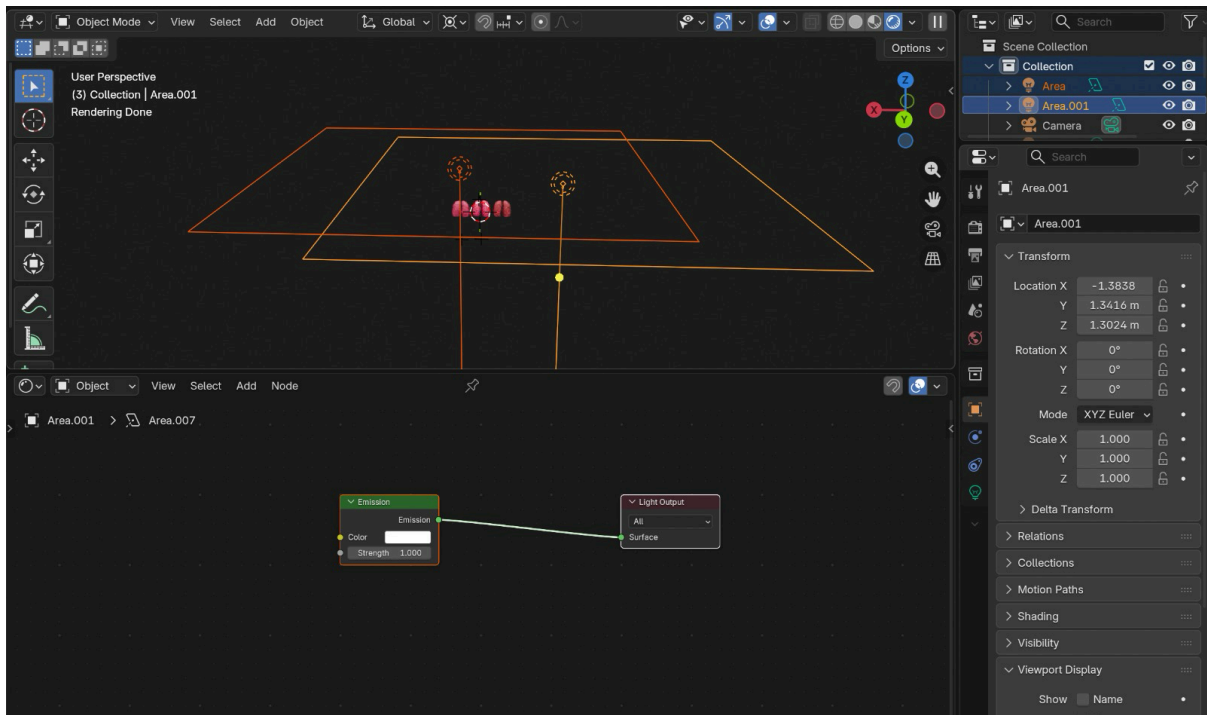
The earliest stage of our technical development explored the possibility of using Blender, a professional 3D modelling and rendering application, to produce the visual representation of the lung model. The initial motivation for this choice was visual realism. Blender's physically based rendering engine, advanced material system, and intricate lighting tools offered the distinct potential to produce organ visualisations of a quality significantly higher than what could be achieved through a purely programmatic rendering solution alone. Initial experimentation with Blender demonstrated that its material and shading capabilities were indeed capable of producing compelling anatomical renders. Therefore, the team invested considerable time in exploring how disease related visual changes could be represented dynamically through material property modifications.

This initial approach, however, encountered a series of fundamental problems that ultimately made it completely unsuitable for the project's strict requirements. The first major issue was material export reliability. Blender's material system uses a complex, node based architecture that simply does not export cleanly to standard interchange formats such as GLB or OBJ when intricate shader networks are involved. As a consequence, materials that appeared perfectly correct in the Blender viewport frequently failed to render as expected when the exported model was loaded into a Python environment. The second related problem was environmental consistency. The rendering behaviour of Blender materials is highly dependent on the specific version of the software and the rendering engine in use. This dependency made it incredibly difficult to guarantee consistent visual results across different user machines or operating environments.

The third, and most significant problem, centered on reproducibility. The proposed workflow required manual intervention at multiple developmental stages, such as adjusting material parameters, manually re-exporting models, and visually verifying the outputs. This fragmentation made it fundamentally unsuitable for an automated system designed to process multiple CT cases rapidly and produce consistent, repeatable results. After a careful and objective assessment of these limitations against the project's core requirements of transparency, reproducibility, and pipeline integration, the team made the decision to abandon the Blender approach entirely. Instead, the project transitioned to a fully programmatic

rendering pipeline using PyVista and Trimesh. This crucial design decision is discussed further in Section 3.5 in the context of risk mitigation.





3.3.2 CT Image Processing Pipeline

The CT image processing pipeline was developed as a standalone module, initially implemented in `ct_step3_lung_mask.py`, before being refactored into the reusable `extract_ct_features()` function that forms the first stage of the integrated pipeline. The pipeline operates as follows.

Each input CT image is first converted to grayscale using OpenCV's `cvtColor` function, producing a single-channel intensity image in which the brightness of each pixel reflects the radiodensity of the corresponding tissue. A Gaussian blur with a 5×5 kernel is applied to reduce acquisition noise and smooth minor intensity variations that could interfere with subsequent thresholding.

Otsu's method is then applied in inverse mode to automatically determine the optimal intensity threshold that separates the dark lung regions from the surrounding brighter soft tissue. The resulting binary image is passed to OpenCV's connected component analysis function, which identifies all contiguous regions of foreground pixels and computes their geometric property area, bounding box dimensions, and centroid coordinates.

A sequence of anatomically motivated filters is then applied to retain only components that are consistent with lung tissue: components must exceed a minimum area threshold to exclude noise, must fall below a maximum area threshold to exclude the background and chest wall, must not touch the image border, and must be located within the central thoracic region of the image both vertically and horizontally. Morphological closing and opening operations are then applied to fill small holes in the segmentation mask and remove minor

artefacts. Finally, only the two largest remaining components are retained, corresponding to the left and right lungs.

From the resulting lung mask, three quantitative signals are extracted. Severity is computed as the mean pixel intensity of the masked lung region, normalised to the range $[0, 1]$. Texture is computed as the standard deviation of pixel intensities within the mask. The high-density ratio is computed as the proportion of lung pixels whose intensity exceeds 0.6, a threshold that corresponds to denser, potentially fibrotic tissue regions.

This pipeline was applied to eight CT cases sourced from Radiopaedia, producing the feature values summarised in Table 1 of Section 4.

3.3.3 Reference Lung Model and 3D Setup

Rather than constructing a lung mesh from scratch a process that would have required either manual modelling or full volumetric CT reconstruction, both of which were beyond the scope of this project, the team used a pre-built anatomical lung model in GLB format obtained from the Human Atlas reference organs API. This model provided a consistent, anatomically accurate geometric substrate that had been pre-validated for structural correctness.

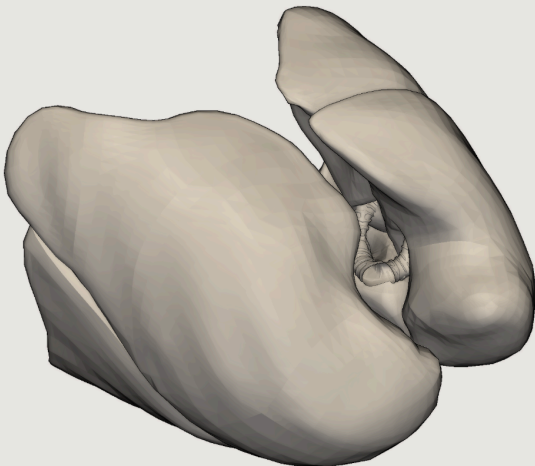
The model is loaded using the Trimesh library, which handles the GLB format and provides access to vertex coordinates and face indices. The geometry is then converted to a PyVista PolyData object, which supports the mesh manipulation operations required for disease simulation. A smoothing pass using 30 iterations of Laplacian smoothing with a relaxation factor of 0.015 is applied to remove minor surface irregularities and ensure stable geometry for subsequent deformation. Surface normals are computed and stored, as they are required for the normal-direction displacement operations used in the roughness and indentation components of the deformation model.

Anatomical weighting fields are then precomputed from the vertex geometry. A subpleural weight field is derived from the normalised radial distance of each vertex from the mesh centroid, concentrating disease influence at the outer surface of the lung. A basal weight field is derived from the vertical position of each vertex, concentrating disease influence in the lower lung regions. The combined weight field, which is the product of these two components, reflects the known spatial distribution of IPF, predominantly subpleural and basal, and ensures that all disease simulation effects are anatomically localised rather than uniformly distributed across the mesh.

Stage: Healthy

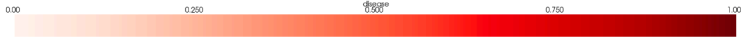
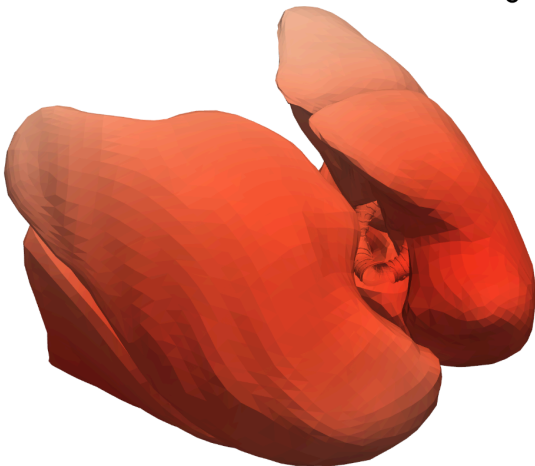
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Disease Progression



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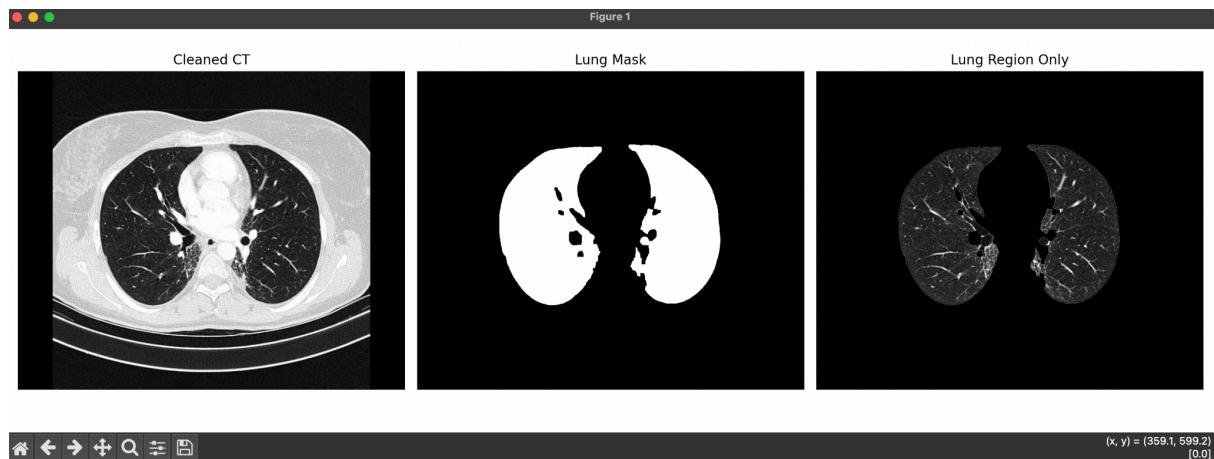
Disease Progression



3.3.4 Disease Simulation Model

The disease simulation model generates a per-vertex disease intensity field that drives all visual and structural transformations. Ten lesion centres are placed deterministically using a fixed random seed, selected from the subset of vertices with the highest anatomical weight values. For each vertex, the disease field value is computed as the maximum Gaussian influence from any lesion centre, with the Gaussian radius scaled by the lesion spread parameter derived from the CT high-density ratio. Additional spatial variation is introduced through a sinusoidal blob field and a cluster field, both of which modulate the disease intensity to produce an organic, heterogeneous distribution rather than a uniform expansion from discrete centres.

The progression parameter t , which the user controls through the interactive slider scales the intensity of all disease effects through a smooth interpolation function. Crucially, the upper bound of this interpolation is not fixed but is determined by the CT-derived parameters for the current case. A case with high severity will reach full colour saturation at a lower slider value than a mild case; a case with high HDR will show more extensive lesion spread; a case with high texture will exhibit more pronounced surface roughness. This coupling between the CT signals and the simulation parameters is what distinguishes this system from a generic animation; the visual output is a direct, deterministic function of the imaging evidence.



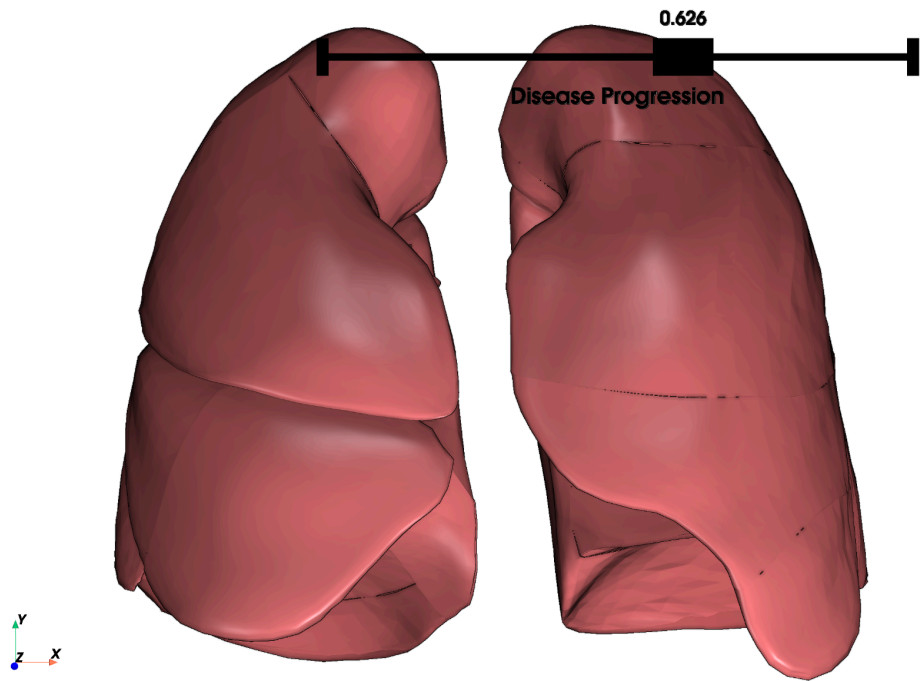
3.3.5 Structural Deformation

Four geometrically distinct deformation effects are applied to the mesh vertices in proportion to the local disease field value. Volume loss is modelled as a contraction of each affected vertex toward the mesh centroid, reflecting the reduction in lung volume associated with progressive fibrosis. Surface indentation is modelled as inward displacement along the vertex normal, reflecting the retraction of the pleural surface in fibrotic regions. Surface roughness is modelled as displacement along the vertex normal in proportion to a high-frequency sinusoidal pattern, reflecting the irregular, textured surface morphology of fibrotic tissue. Basal collapse is modelled as downward displacement in proportion to the basal weight field, reflecting the known tendency of IPF to cause greater volume loss in the lower lobes.

All deformation magnitudes are scaled by the deformation strength parameter derived from the CT progression score, ensuring that the structural changes are proportional to the severity

of the case. A higher threshold for activating deformation and a steeper local falloff curve ensure that only vertices with substantial disease field values are deformed, preventing the geometric distortions that were observed in earlier iterations of the model when deformation occurs.

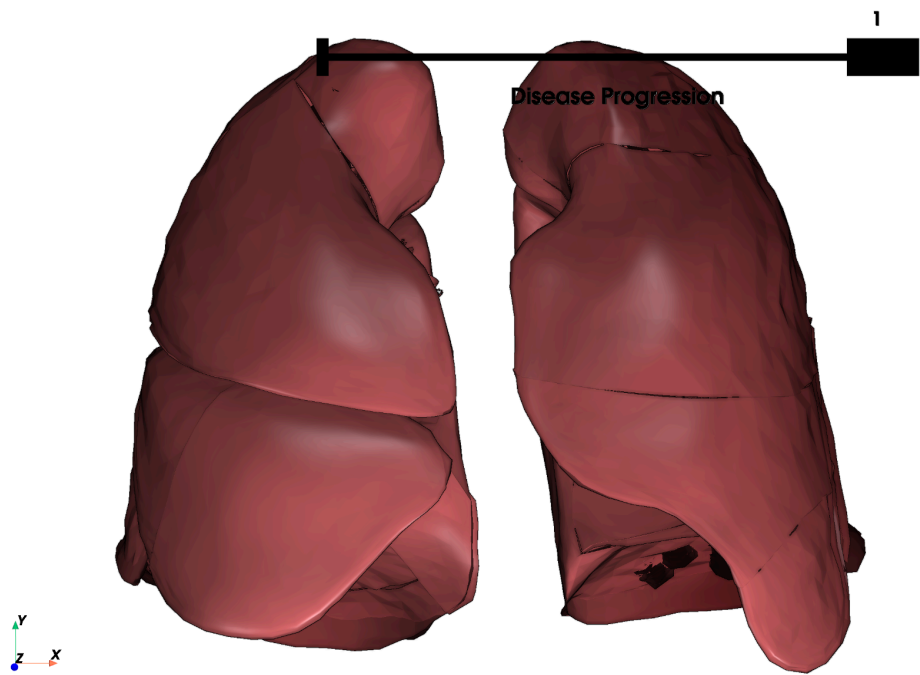
Stage: Moderate Disease



Moderate disease: abnormal regions spread with visible tissue deformation.

was applied too broadly.

Stage: Advanced Disease

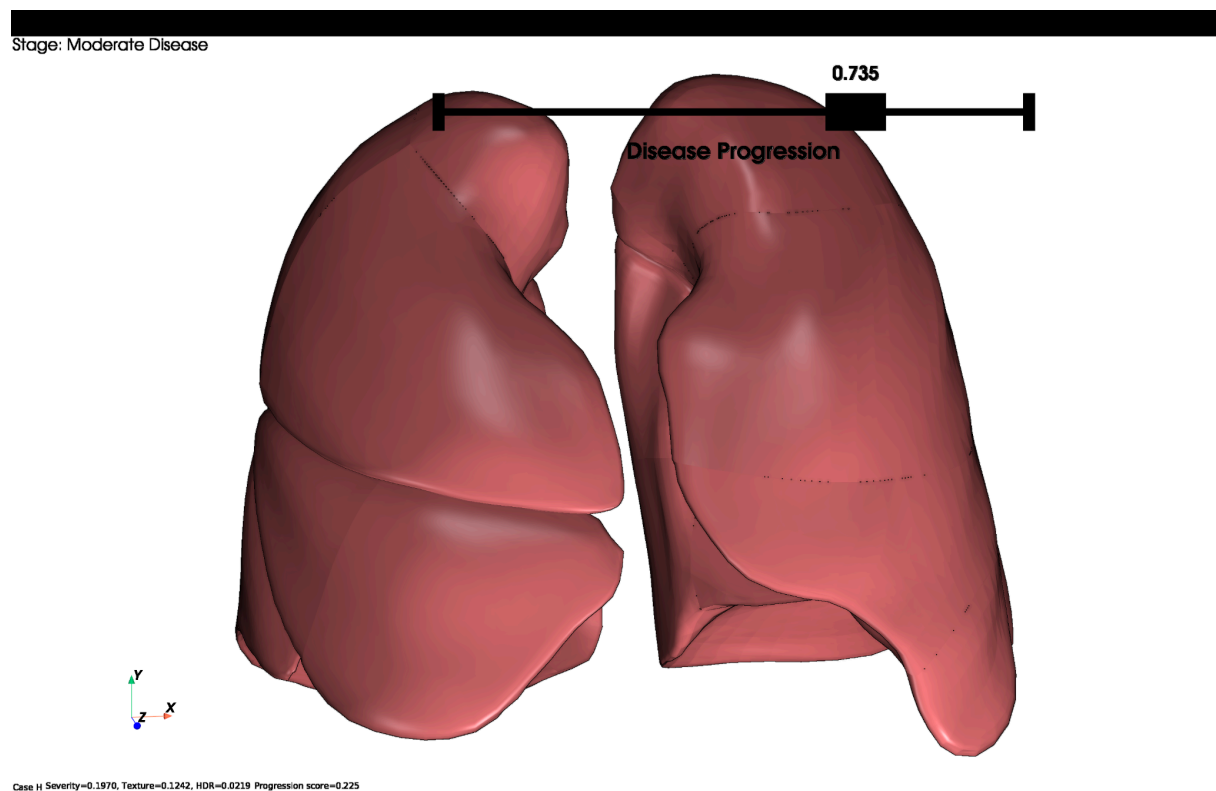


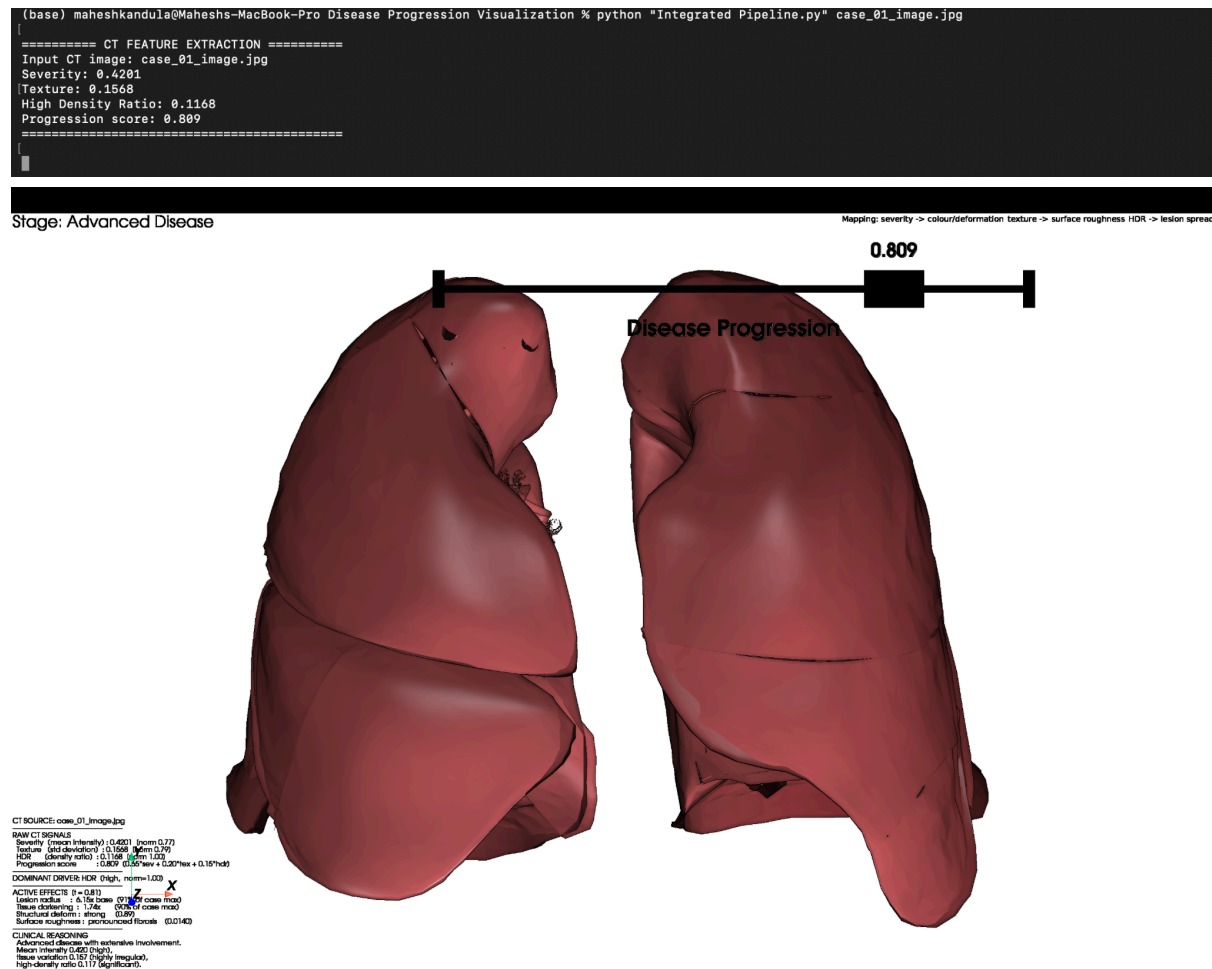
Advanced disease: widespread density change, volume loss, and roughened morphology.

3.3.6 Visual Rendering and the Explanation Panel

The interactive viewer is built using PyVista's Plotter class, configured with a three-point lighting setup key, fill, and rim lights that provides depth and surface clarity. Smooth shading, specular highlights, and ambient occlusion are applied to give the mesh a realistic tissue-like appearance. The background is set to white to maximise contrast with the tissue colours.

The live explanation panel is rendered as text overlaid on the lower-left region of the viewer window. It displays the CT source image filename, the three raw CT signals with their normalised values, the dominant signal driver for the current case, the active visual effects at the current slider value including lesion radius multiplier, tissue darkening level, deformation state, and surface roughness and a plain-language clinical reasoning statement derived from the case's progression score and signal profile. This panel updates every time the slider moves, providing the user with a continuous, real-time account of why the model looks the way it does through the progression.

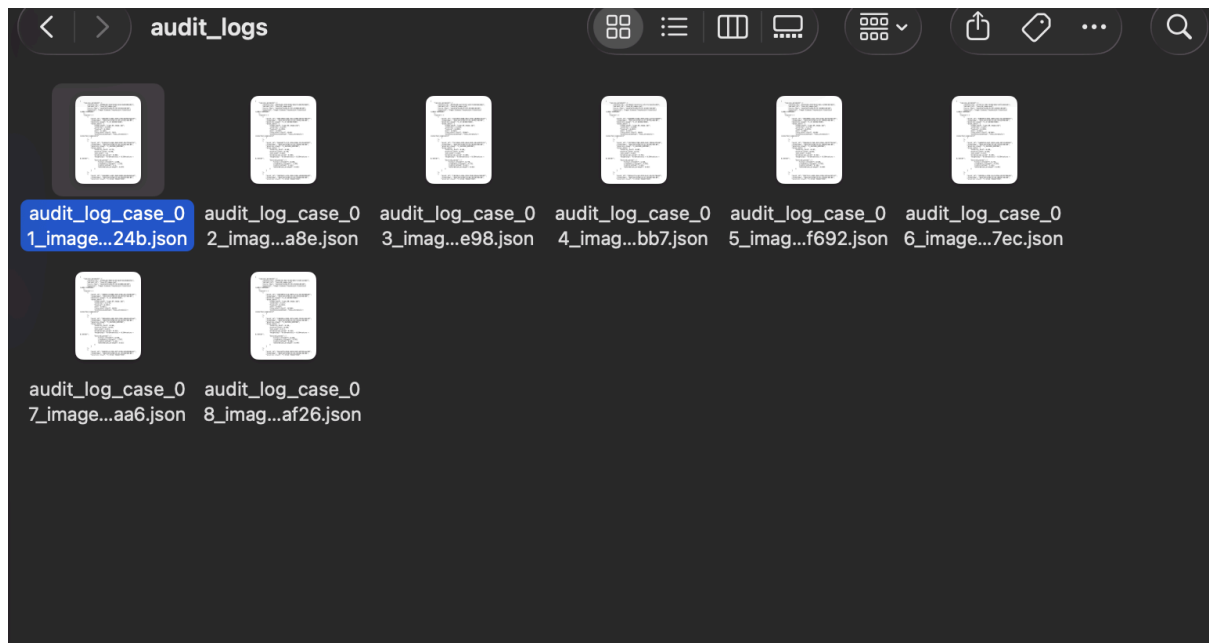




3.3.7 Audit Logging System

The ProgressionAuditLogger class, initially designed by team member David Kong and subsequently integrated and wired to real pipeline values by Mahesh Kandula, provides a structured record of every decision made by the pipeline for each case. Each session produces a JSON file containing four categories of logged events. Stage 1 records the CT extraction results, the image path, the three raw signal values, the lung pixel count, and the extraction method. Stage 2 records the feature mapping results, the normalised signal values, the progression score, the weighting formula, and the four derived visualisation parameters. Stage 3 records stage transitions logged each time the disease stage changes as the slider moves, capturing the slider value, the stage label, and the active effect multipliers with a plain-language clinical justification. Stage 4 records vertex modification details the number of affected vertices, the deformation component magnitudes, the mesh region, and a clinical justification for the geometric changes applied.

These logs are saved to an audit_logs/ directory on exit when the --save_audit flag is passed at runtime. Eight audit logs, one per clinical case, were produced and are presented alongside the visualisation results in Section 4.



3.4 Reframing the Problem

The original conception of this project, as described in the preliminary progress report, envisioned a Large Language Model as the primary driver of 3D mesh transformations. This was intended as a system in which clinical text descriptions would be interpreted by a model such as ClinicalBERT and translated directly into geometric changes in the lung model. Early in the development process, however, it became clear that this approach, while conceptually compelling, introduced a level of non-determinism that was fundamentally incompatible with the project's clinical accountability requirement. For instance, an LLM that produces slightly different parameter values each time it processes the exact same clinical text cannot form the basis of a system whose outputs are claimed to be traceable and reproducible.

The core problem was therefore thoroughly reframed. Rather than using a volatile LLM to drive the 3D model, the project would use imaging-derived signals as the primary control mechanism. Consequently, the LLM component was reconsidered as a future extension reserved for generating natural language explanations or approximate visualisation parameters from clinical text.

This major reframing had two important consequences for the technical architecture. First, it forced a much more rigorous engagement with the CT signal extraction problem. Instead of treating imaging signals as a secondary input, the team invested significant effort in building a robust, validated segmentation and feature extraction pipeline. Second, it produced a finished system whose outputs are genuinely deterministic and auditable. As a result, this represented a much stronger and more defensible clinical outcome than what the original LLM-driven approach would have yielded.

3.5 Timeline

Week	Focus	Responsibility
1-3	Problem definition, organ And disease selection, literature review and medical domain research	David
3-5	Blender exploration for visual realism, including materials set up, lightning experiments. and export testing	Damaranath, Mahesh
4-5	CT image loading and pre-processing(ct_step1_load_image.py)	Mahesh
5-6	Lung segmentation and feature extraction(ct_step 3_lung_mask.py) for preliminary submission	Mahesh
5-6	Development of first inactive 3-D prototype with progression slider (progression_engine.py)	Mahesh
6-7	Evaluation of Blender limitations and transition decision to pyVista/trimesh pipeline	Damaranath
7-8	Full prototype development with anatomical waiting, Deformation logic, and tissue colouring(prototype.PY) for preliminary submission	Mahesh, damaranath
8-9	CT driven visualiser using hard coded patient case values(ct_driven_visualiser.py)	Mahesh
9	Keyboard paste case switching implementation(keyboard_switching.py)	Mahesh
9	Audit log class design and Json audit structure	David
9-10	ClinicalBERT text Extraction prototype and structured Jason output generation	Chirag srinivasamurthy
10	Argparse-based single case execution	Mahesh

	pipeline with live CT extraction support	
10-11	Integration of audit logger pipeline outputs extracted values	Mahesh, Damaranath
11	Replacement of labels with live explanation panel and interpretation display	Mahesh
11	Stability improvements for hives severe deformation cases	Mahesh, Damaranath
11-12	Final integrated pipeline development and validation across all 8 CT cases (integrated pipeline.PY)	Mahesh,
12	Final report writing, slide preparation and presentation delivery	All Team members

3.6 Risk Mitigation

Blender Integration Failure. The most significant technical risk that materialised during the project was the failure of the Blender-based rendering approach. As described in Section 3.3.1, material export inconsistencies, environment dependency, and the requirement for manual intervention made Blender unsuitable for an automated, reproducible pipeline. The fallback strategy transitioning to a fully programmatic PyVista and Trimesh pipeline was identified and executed promptly. While this transition resulted in lower visual realism compared to what Blender could have produced, it delivered a system with full programmatic control, consistent cross-environment behaviour, and seamless integration with the CT processing pipeline. In retrospect, the PyVista approach was better aligned with the project's core requirements from the outset, and the Blender exploration, while ultimately unproductive, provided a useful basis for understanding the trade-offs involved.

CT Segmentation Sensitivity. The Otsu thresholding approach proved sensitive to image quality, contrast, and acquisition parameters across the eight CT cases. Several cases required careful tuning of the connected component filtering thresholds to produce clean segmentation masks. The mitigation strategy was to implement a series of anatomically motivated geometric filters area bounds, boundary proximity checks, and centroid location constraints that together provide robust segmentation across images of varying quality without requiring case-specific parameter adjustments.

Mesh Deformation Instability. Early versions of the deformation model used parameter values that caused geometric distortion and mesh tearing at high progression values for severe cases, particularly Case 01 which had the highest severity score in the cohort. The mitigation was to recalibrate the deformation constants reducing volume loss, roughness, and collapse strengths and to raise the disease field threshold below which deformation is not

applied. This produced a stable, visually coherent model across the full progression range for all eight cases.

LLM Integration Scope. The original plan to integrate a ClinicalBERT-based text extraction component into the pipeline was descoped following the problem reframing described in Section 3.4. The mitigation was to document the intended architecture clearly including Chirag Srinivasamurthy's prototype JSON output format so that the LLM component can be integrated in future work without requiring changes to the core pipeline.

Time Constraints. Given the volume of technical work required to build and validate the integrated pipeline, not all planned features were fully implemented. The LLM component, multi-slice CT processing, and a comparative validation against clinical ground truth were identified as beyond the scope of the current prototype. These are discussed as future directions in Section 6.

4. Results, Outcomes and Analysis

4.1 CT Image Processing and Feature Extraction

The CT processing pipeline was successfully applied to all eight clinical cases sourced from Radiopaedia, producing clean lung segmentation masks and three quantitative disease signals for each case. The pipeline operated without case-specific parameter adjustments; the same segmentation logic, filter thresholds, and morphological operations were applied uniformly across all eight images, which varied in format (JPG and PNG), resolution, acquisition parameters, and visual characteristics.

Table 4.1 presents the extracted feature values for all eight cases, computed live from the CT images by the integrated pipeline at runtime. These values supersede the manually recorded values reported in the preliminary submission, which were computed by running the segmentation script individually on each image and transcribing the outputs by hand. The final pipeline eliminates this manual step entirely; feature extraction is now an automated first stage of every run.

Table 4.1 — Live CT Feature Extraction Results Across All Eight Cases

Case	Image file	Severity(Mean Intensity)	Texture(St d Dev)	HDR(density ratio)	lung pixels	progress ion score
Case 01	case_01_image.jpg	0.4201	0.1568	0.1168	7937	0.809
Case 02	case_02_image.jpg	0.3537	0.1556	0.0766	31462	0.668
Case 03	case_03_image.jpg	0.3255	0.0985	0.0218	170647	0.409
Case 04	case_04_image.jpg	0.4603	0.0932	0.0774	28621	0.754
Case 05	case_05_image.jpg	0.2713	0.0745	0.0097	41430	0.242
Case 06	case_06_image.jpg	0.3070	0.1209	0.0324	441503	0.436
Case 07	case_07_image.jpg	0.2772	0.1357	0.0411	52972	0.423
Case 08	case_08_image.jpg	0.1970	0.1242	0.0219	121590	0.218

Table 4.1 — CT feature values extracted live from each case image by the integrated pipeline. Progression score computed as $0.65 \times \text{severity_norm} + 0.20 \times \text{texture_norm} + 0.15 \times \text{hdr_norm}$.

Several observations are worth noting from these clinical results. For instance, Case 01 and Case 04 represent the most severe cases in the cohort, exhibiting the highest severity values of 0.4201 and 0.4603 respectively, alongside the highest high density ratio values. Consequently, these metrics produce elevated progression scores of 0.809 and 0.754. These cases correspond directly to the most advanced disease stages within the visualiser.

Conversely, Case 08 is the mildest case in the study, demonstrating the lowest severity value of 0.1970 and a progression score of 0.218. Therefore, this specific dataset produces an early stage visualisation characterised by minimal deformation and a smooth surface texture.

Often, a broad distribution of data is necessary to prove overall system utility. The range of progression scores across this cohort, spanning from 0.218 to 0.809, is completely sufficient to produce clearly differentiated visual outputs at each end of the spectrum. Ultimately, this wide variance validates that the three extracted signals together provide meaningful discrimination across diverse disease severity levels.

The lung pixel counts vary considerably across the patient cases, ranging from 7,937 for Case 01 to 441,503 for Case 06. This wide variation reflects clear differences in image resolution, field of view, and the specific extent to which lung tissue is visible in the particular CT slice used.

Notably, Case 01's exceptionally low pixel count did not impair the feature extraction process whatsoever. The severity and high density ratio values remain the highest in the cohort, which is entirely consistent with the heavily diseased lung tissue visible in that specific CT image. In this instance, dense fibrotic regions occupy a massive proportion of the visible lung area, allowing the pipeline to calculate an accurate progression profile despite the constrained image dimensions.

4.2 Three-Dimensional Visualisation Results

The integrated pipeline was run for all eight cases, producing interactive 3D visualisations in which the lung mesh is driven directly by the live-extracted CT signals. Figures 4.1 through 4.8 present the visualiser output for each case at the slider value corresponding to the case's progression score, the point at which the 3D model most directly reflects the CT imaging evidence.

Case 01 — Advanced Disease (progression score: 0.809, $t = 0.809$)

Case 01 represents the most severe case in the cohort. The visualiser correctly assigns it to the Advanced Disease stage. The mesh exhibits strong darkening toward deep fibrotic red, substantial volume contraction, and surface roughness consistent with pronounced fibrosis. The explanation panel identifies HDR as the dominant driver (norm = 1.00), reflecting that Case 01 has the highest high-density ratio in the cohort. The lesion radius is scaled to 6.15 times the base radius, and structural deformation is at 0.89 the highest deformation value across the cohort. The audit log records 264,370 affected vertices with a volume loss strength of 0.142, confirming the extent of geometric transformation.

Case 02 — Moderate Disease (progression score: 0.668, $t = 0.668$)

Case 02 is assigned to the Moderate Disease stage. The mesh shows significant colour darkening and visible volumetric changes, particularly in the lower lobes, consistent with the basal weighting of the deformation model. The dominant driver is HDR (norm = 0.90), with texture also elevated (norm = 0.78), producing a moderately rough surface. Lesion radius is 4.79 times the base radius and deformation is at 0.63. The audit log records volume loss strength of 0.101 and basal collapse of 0.057, reflecting substantial but controlled geometric change.

Case 03 — Early Disease (progression score: 0.409, t = 0.409)

Case 03 sits in the early disease range. The mesh retains a largely healthy pink colouration with mild reddening in the subpleural and basal regions. The dominant driver is Severity (norm = 0.50), and the surface roughness is smooth consistent with the low texture norm of 0.26. Lesion radius is 1.48 times the base radius, deformation is mild at 0.23, and the audit log confirms roughness strength of only 0.002 the explanation panel correctly labels this as smooth. This case demonstrates the system's ability to produce a visually restrained output for a case with moderate but not severe imaging features.

Case 04 — Advanced Disease (progression score: 0.754, t = 0.754)

Case 04 is one of two advanced disease cases. Despite having a higher raw severity value (0.4603) than Case 01 (0.4201), its progression score of 0.754 is slightly lower because its texture norm is lower (0.21 vs 0.79). This highlights how the weighted scoring system produces nuanced differentiation two cases with similar raw severity can produce different progression profiles depending on their texture and HDR characteristics. The visualiser correctly captures this: Case 04 shows strong deformation and colour change but with smoother surface texture than Case 01. The dominant driver is again HDR (norm = 0.90), with a lesion radius of 5.41 times the base and deformation at 0.79.

Case 05 — Early Disease (progression score: 0.242, t = 0.242)

Case 05 is the mildest case after Case 08. The progression score of 0.242 produces a visualisation that is visually close to healthy light pink colouration, very minimal deformation (0.07), and a lesion radius below the base radius (0.93x). The dominant driver is Severity at norm = 0.35, which is classified as low. The explanation panel correctly states near-healthy or very mild abnormality. This case is useful for demonstrating the healthy end of the spectrum and provides a visual anchor for comparison with the more severe cases.

Case 06 — Early Disease (progression score: 0.436, t = 0.436)

Case 06 is notable for having the largest lung pixel count in the cohort (441,503), reflecting its high-resolution PNG format and the large extent of visible lung tissue in the image. Despite this, the extracted signals are moderate, producing an early disease classification. The dominant driver for Case 06 is Texture (norm = 0.46), reflecting relatively high tissue irregularity compared to its severity. The surface roughness activates at a mild fibrotic texture level (0.003), and the lesion radius is 1.82 times the base. The visualiser produces a model with mild colour change and early-stage deformation consistent with the imaging evidence.

Case 07 — Early Disease (progression score: 0.423, t = 0.423)

While Case 07 is similar in severity to Case 06, it presents a entirely different diagnostic signal profile. For instance, structural texture is the dominant driver with a normalised value of 0.60, directly reflecting the irregular tissue structure visible in the underlying CT image. Consequently, the surface roughness activates at a mild fibrotic level of 0.003, while the overall structural deformation remains mild at 0.24. The explanation panel correctly identifies this state as early stage disease characterized by localised changes, highlighting elevated

tissue variation as the distinguishing clinical feature. Furthermore, the generated audit log confirms 264,370 affected vertices with strictly controlled deformation components.

Case 08 — Early Disease (progression score: 0.218, $t = 0.218$)

Case 08 is the mildest case in the cohort and was the primary development case used throughout the project its values closely match the manually recorded Case H values from the preliminary submission (severity 0.197 vs 0.1970, texture 0.124 vs 0.1242, HDR 0.022 vs 0.0219), confirming that the live extraction pipeline is consistent with the manual pipeline used during development. The dominant driver is Texture (norm = 0.49), the deformation is minimal (0.05), and the surface is smooth. The clinical reasoning panel correctly classifies this as near-healthy or very mild abnormality.

4.3 Audit Log Validation

All eight cases produced structured JSON audit logs when run with the `--save_audit` flag. Each log contains four pipeline stages: CT extraction, feature mapping, stage transition, and vertex modification. Table 4.2 summarises the key audit log values across all cases.

Table 4.2 — Audit Log Summary Across All Eight Cases

Case	stage assigned	lesion Radius mult	intensity mult	deforme mult	affected vertices	volume loss
Case 01	Advanced disease	6.152X	1.739X	0.885	264370	0.142
Case 02	Moderate disease	4.794X	1.218X	0.633	264370	0.101
Case 03	Early disease	1.475X	0.602X	0.226	264326	0.036
Case 04	Advanced disease	5.408X	1.794X	0.788	264370	0.126
Case 05	Early disease	0.931X	0.266X	0.069	258641	0.011
Case 06	Early disease	1.822X	0.621X	0.259	264370	0.041
Case 07	Early disease	1.986X	0.545X	0.244	264370	0.039
Case 08	Early	0.993X	0.195X	0.054	254901	0.009

	disease					
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Table 4.2 - Audit log summary values for all eight cases. Each value is traceable to the CT signals extracted in Stage 1 of the pipeline.

The audit logs confirm that the pipeline behaves deterministically; running the same case twice produces identical values. They also confirm that the transformation parameters scale monotonically with disease severity: progression from Case 08 through to Case 01 shows a consistent increase in lesion radius multiplier, intensity multiplier, deformation multiplier, and volume loss strength. This monotonic scaling is an important validation of the mapping function; it demonstrates that the system correctly encodes the ordering of disease severity from the CT signals into the visual and structural output.

The affected vertex count is consistent across most cases at approximately 264,370, which represents the total number of vertices in the smoothed lung mesh. The slight reduction for Case 05 (258,641) and Case 08 (254,901) reflects the higher disease field threshold in mild cases, a smaller proportion of vertices exceeding the threshold required to activate deformation effects, which is the correct behaviour.

4.4 Live Explanation Panel Analysis

The live explanation panel renders six categories of information in the bottom-left of the viewer window, updating in real time as the slider moves. Across all eight cases, the panel correctly identifies the dominant CT signal, quantifies the active visual effects at the current slider value, and provides a plain-language clinical reasoning statement consistent with the case's progression score and signal profile.

Several examples demonstrate the panel's discriminative capability. For Case 01, the panel identifies HDR as the dominant driver and states "Advanced disease with extensive involvement" appropriate for a case with the highest progression score and the highest HDR in the cohort. For Case 05, the panel identifies severity as the dominant driver at low strength (norm = 0.35) and states "Near-healthy or very mild abnormality" appropriate for the second-mildest case. For Case 07, the panel identifies Texture as the dominant driver at moderate strength (norm = 0.60) and produces a clinical reasoning statement that correctly notes elevated tissue variation, the distinguishing feature of that case relative to its neighbours in the severity ranking.

This per-case differentiation in the explanation panel demonstrates that the system is not producing generic outputs; the reasoning adapts to the specific signal profile of each CT image, providing the user with a meaningful account of why the model looks the way it does for that particular case.

4.5 Discussion

4.5.1 Alignment with Project Aims

The three core aims established in Section 1.1 have all been achieved. The deterministic imaging-to-3D pipeline is fully implemented and validated across eight real CT cases. Clinical accountability is demonstrated through the audit logging system, which provides a permanent, per-case record linking every visual transformation to the CT signal that caused it. The interactive visualisation tool communicates disease progression through a combination of 3D mesh transformation, colour mapping, and live explanatory text that is accessible to non-specialist users while retaining technical rigour.

4.5.2 Limitations and Weaknesses

Several limitations of the current system are worth acknowledging honestly. The CT feature extraction operates on single 2D image slices rather than full volumetric CT stacks, which means the extracted signals represent only one cross-section of the lung at a single axial level. A more rigorous approach would process multiple slices across the full lung volume and aggregate features across the three-dimensional extent of the disease. The current approach was a deliberate scope decision for the prototype, prioritising pipeline integration and interpretability over extraction completeness.

The segmentation pipeline, while robust across the eight test cases, relies on anatomically motivated heuristics that may not generalise to CT images with unusual acquisition parameters, significant artefacts, or atypical patient anatomy. Case 01's relatively low lung pixel count (7,937 compared to Case 06's 441,503) suggests that the segmentation captured only a subset of the visible lung tissue for that image, which may affect the reliability of the extracted features. Future work should incorporate more robust segmentation methods, potentially including deep learning-based approaches such as U-Net architectures trained on annotated lung CT datasets.

The 3D model itself is a single reference mesh from the Human Atlas API, representing a healthy adult lung. It does not vary between patients, which means the deformation model is applied to a standard anatomical template rather than to a patient-specific geometry. This is appropriate for a prototype and for educational visualisation, but it limits the clinical applicability of the system; a patient's actual lung geometry may differ substantially from the reference model, and disease effects that are highly specific to that geometry cannot be represented.

4.5.3 Retrospective Assessment

With the benefit of hindsight, the decision to prioritise imaging-derived signals over LLM-driven text input as the primary control mechanism was correct. The imaging pipeline produces consistent, reproducible, quantitatively grounded outputs that directly satisfy the clinical accountability requirement. An LLM-driven approach would have required substantially more infrastructure model hosting, prompt engineering, output validation, and hallucination mitigation for a component that, in the context of this prototype, adds expressive power at the cost of determinism.

The Blender exploration, while ultimately unproductive, was not wasted time. It forced a clear articulation of what the project actually required from its rendering pipeline: programmatic control, reproducibility, and pipeline integration and made the case for PyVista and Trimesh compelling and well-reasoned rather than a default choice. The resulting system is more aligned with the project's core values precisely because the Blender alternative was seriously considered and found wanting.

4.5.4 New Research Directions

Several new research directions emerged during the course of this project. The most technically interesting is the question of how to extend the signal-to-parameter mapping to accommodate text-based clinical inputs. Chirag Srinivasamurthy's prototype JSON output — `{"density": "increased", "region": "apico_basal_lung", "scarring": "present", "severity": "advanced"}` — provides a concrete specification of the structured format that a ClinicalBERT extraction layer could produce, and which could be mapped to the existing visualisation parameters using a rule-based translation layer. This extension would not require changes to the core pipeline and could be implemented as an optional input mode alongside the existing CT image input.

A second direction is the development of a comparative validation framework, a method for quantifying how well the 3D visualisation corresponds to the clinical reality of each case. This would require annotated ground truth from radiologists or pulmonologists, specifying the expected stage, dominant disease region, and approximate extent of fibrotic involvement for each case. Such validation would elevate the system from a prototype to a clinically credible tool and is the most important prerequisite for any real-world deployment.

5. Individual Contributions

Team member: David Kong (47458992)

Contribution:

My contribution to this group project has been a mix of project logistics, technical research, report writing & production and some technical development.

A key focus of mine was undertaking the bulk of the research in both the medical domain and data science/machine learning space to facilitate the formulation of the research topic that included the research topic/question, the objectives, as well as the goals. In addition to this, the research I conducted was to support the on-going development of the technical solution by grounding the technical decision at each step in evidence based methodology.

In the initial stages of the project, the research I conducted explored the validity of different organs as a candidate for use in the project prototype, as well as the associated disease diagnosis we would focus on. Given project time constraints we faced, selecting the right organ and disease to use was imperative as there was only scope to select one of each. Furthermore, research I conducted examined the current use case of medical imaging in Australia and the wider world as well as the pain points currently being felt within the status

quo. In undertaking this medical research not only were academic papers and textbooks examined but consulted directly with a Staff Specialist Physician (Consultant Doctor) practicing at Westmead Hospital, a major NSW tertiary hospital.

In the data science & machine learning space, I examined the literature that pertained to the technologies we may choose to employ and their applicability to solving the decomposed reached sub-problems we faced. This included ClinicalBERT, 3D mesh formulations including algorithms such as Marching Cubes, and agentic AI models.

In the technical development space, I developed the audit logging script used within our prototype that tracked every decision made by the tool and outputted the results into an easily accessible raw JSON file for review. These logs act as the key accountability competent of our project and helps to ensure that any visualisations made by the tool are evidenced back and auditable.

In addition, I have been responsible for a significant portion of the report writing including the Abstract, Introduction and Literature review as well taking charge of the overall report formatting, editing and presentation slide development.

Finally, led the group from an administrative standpoint and organised the team meetings, meetings with supervisors, maintained the timeline of deliverables and facilitated our group discussions.

Team Member: Mahesh Sai Kandula

Contribution:

My contribution was focused on the complete technical implementation of the system, covering the CT image processing pipeline, the 3D disease progression model, and the final integrated pipeline.

I built the CT processing pipeline using OpenCV, implementing greyscale conversion, Gaussian smoothing, Otsu thresholding, connected component analysis, and morphological operations to segment lung regions from CT images. From the segmented region I extracted three key signals — severity, texture, and high-density ratio. These were tested across all eight Radiopaedia cases to validate consistency.

For the 3D visualisation I developed the full disease simulation model including Gaussian-based lesion propagation, anatomical weighting concentrated in the subpleural and basal regions, four-component structural deformation covering volume loss, surface indentation, surface roughness, and basal collapse, and a tissue colour gradient from healthy pink to advanced fibrotic red. I built the interactive PyVista viewer with a progression slider across multiple iterations — from the early prototype through to the final system.

For the final pipeline I integrated the CT processing and 3D visualiser into a single end-to-end system using argparse for command-line operation, with live feature extraction replacing all hardcoded values. I added the `--show_ct` segmentation preview flag, built the

live explanation panel showing CT signals, dominant driver, active effects, and clinical reasoning in real time, and integrated David Kong's audit logger class with real pipeline values. I also resolved mesh deformation instability that caused geometry tearing at high progression values for severe cases.

The outcome is a fully integrated, deterministic pipeline that takes any CT image as input, extracts signals live, drives the 3D model, explains every visual decision on screen, and saves a structured JSON audit trail on exit.

Future Contribution:

Going forward I would focus on extending the pipeline to process full volumetric CT stacks rather than single slices, integrating Chirag's ClinicalBERT text extraction component as an optional input mode, and developing a clinical validation framework with radiologist ground truth assessments to evaluate how well the system's outputs correspond to real clinical findings.

Team member [Damaranath kokkula-48479373]:

Contribution so far:

Contributed to the research and interpretability aspects of the project by investigating disease progression visualisation challenges, explainable AI concepts, and the broader significance of interpretable medical visualisation systems. This research helped shape the project's expected outcomes, risk mitigation strategies, and overall approach to developing a transparent and trustworthy visualisation pipeline.

Participated in the early exploration of Blender-based visualisation techniques to evaluate methods for improving the realism of the three-dimensional lung model. This included experimenting with materials, textures, lighting configurations, and disease-stage representations. Through this process, several workflow, reproducibility, and integration limitations were identified, which contributed to the team's decision to transition to the final PyVista-based approach.

Contributed to the integration and evaluation of the audit logging framework within the processing pipeline. This involved supporting the automatic recording of CT-derived imaging features, progression scores, disease stages, deformation parameters, and generated outputs for each processed case. The audit logging system provides a traceable record linking CT-derived evidence to the final three-dimensional visualisation, supporting the project's interpretability and accountability objectives.

Additionally, contributed to report writing, documentation, project reviews, presentation preparation, and refinement of the project's research, methodology, and interpretability discussions throughout both the preliminary and final stages of development.

Team member Chirag Srinivasamurthy (48467022)

Contribution :

My role was to create an AI that could process the description of lung fibrosis as found in radiology reports and transform it into structured disease features that could be used in the progression engine and in the 3D lung visualisation system.

Rather, I first looked into the application of Large Language Models (LLMs) in health care and biomedical Natural Language Processing (NLP) approaches to clinical text. Based on this review, I decided to use ClinicalBERT as the language model because it was pre-trained using clinical notes and its core was able to capture medical terminology, which is often used in radiology reports and is domain-specific. In addition to the design-first approach of this project, putting trust at the centre, I also studied the reliability, explainability and risk of hallucination of LLMs in healthcare.

Then I developed the module in Python with Jupyter Notebook. ClinicalBERT was added to the pipeline to handle descriptions of radiology text and to provide contextual representations of clinical text. Instead of using free-text generation, I developed and applied a framework for the extraction of pulmonary fibrosis features directly from radiology descriptions, which involved using a set of rules.

Pulmonary fibrosis case studies were obtained from Radiopaedia cases to develop and improve the feature extraction layer. Five important attributes of the disease were used for extraction rules: density abnormalities, anatomical region, scarring, disease pattern and disease severity. To enhance its robustness across different reporting styles, the framework included terms from pulmonary fibrosis, such as "honeycombing", "reticulation", "traction bronchiectasis", "usual interstitial pneumonia (UIP)", "interstitial markings", "architectural distortion", "interlobular septal thickening", "ground-glass opacity", and "fibrosis-related descriptors".

In addition, I created anatomical localisation rules to detect disease distribution patterns including apico-basal lung involvement, lower lung disease, upper lung disease, peripheral lung disease, medial lung disease, bilateral disease, segmental disease and lobar disease. A set of rules for severity classification was built that were able to determine if a disease presentation was early, moderate or advanced, using terms that appeared in pulmonary fibrosis reports.

I designed a structured JSON-based disease representation to enable integration with other components of the project. This format ensures that the disease features extracted from the clinical text are structured in the same way, as well as providing a consistent interface between the clinical text interpretation module and the progression engine.

Team Member: Anil Kumar Varada (48566861)

Contribution

My contribution to this project focused on developing the methodology and CT image analysis pipeline used for pulmonary fibrosis progression visualisation. I worked on processing CT scan images, starting from image preprocessing and lung segmentation through to disease signal extraction and progression mapping.

I implemented image preprocessing techniques such as grayscale conversion, Gaussian filtering, thresholding, and morphological operations to improve image quality for further analysis. I also designed and implemented the lung segmentation process using connected component analysis to identify and isolate lung regions from CT images. After segmentation, I developed the feature extraction module to calculate important disease-related indicators including tissue density severity, structural texture, and high-density ratio (HDR).

In addition, I worked on converting these extracted features into a disease progression parameter that could be used by the visualisation system to represent different stages of pulmonary fibrosis. I also contributed to the audit logging mechanism, which records disease signals, progression values, and visual changes applied to the 3D lung model. Finally, I integrated these components into the overall pipeline to ensure that the visualisation remained explainable, reproducible, and directly linked to the original CT scan data.

Future Contribution:

The methodology can be improved by using larger CT datasets and additional disease indicators to provide more accurate disease analysis. Future work could also include advanced segmentation techniques and full 3D CT volume analysis to enhance visualisation quality.

The explainable AI framework can be further expanded with more detailed audit logs and visual explanations, making the system more useful for clinical and research applications.

6. Conclusion and Future Outlook

6.1 Research Problems Initially Planned

This project was initially conceived as an LLM-driven visualisation engine, a system in which clinical text descriptions of pulmonary fibrosis would be interpreted by a domain-specific language model such as ClinicalBERT and translated directly into coordinate-level transformations of a 3D anatomical lung mesh. The motivation was to address two well-documented failures in existing medical AI visualisation: the lack of determinism, whereby identical inputs produce varying outputs across runs, and the

black-box interpretability gap, whereby no auditable link exists between a visual output and the clinical evidence that produced it. The project brief specifically called for a trust-first transformation approach in which every visual edit is constrained by imaging evidence and produces an auditable trail of what changed, why, and from which imaging cues.

6.2 Problems Actually Addressed

The problems the project ultimately addressed are closely aligned with this original intent, but the primary input mechanism shifted from clinical text to CT imaging signals following the problem reframing described in Section 3.4. Rather than using an LLM to extract parameters from narrative reports, the system extracts three quantitative signals severity, texture, and high-density ratio directly from CT images using a deterministic segmentation and feature extraction pipeline. These signals drive a weighted progression score that controls all visual and structural properties of the 3D mesh deterministically and reproducibly.

This shift did not diminish the project's contribution if anything it strengthened it. The imaging-driven approach produces a tighter, more verifiable link between clinical evidence and visual output than an LLM-driven approach could have provided within the project's time and resource constraints. The audit logging system, which records the full pipeline trail from CT pixel values through to vertex modification counts, provides a concrete implementation of the clinical accountability principle that the project set out to demonstrate.

The Blender rendering approach was also addressed as an implicit problem the team invested time in assessing whether Blender could serve as the rendering platform before concluding that its material export inconsistencies, environment dependencies, and requirement for manual intervention made it incompatible with the project's reproducibility requirements. The transition to PyVista and Trimesh resolved this problem and produced a fully programmatic, pipeline-integrated rendering system.

6.3 Results Achieved

The project achieved all four expected outcomes stated in Section 1.4. A functional three-stage 3D progression model was built and validated across eight real CT cases, demonstrating smooth, continuous transitions from healthy through early and moderate to advanced disease states. A quantitative imaging-to-3D mapping method was implemented, formally specifying how severity drives colour strength and deformation, texture drives surface roughness, and HDR drives lesion spread. A trust-first transformation pipeline was delivered, with every mesh modification traceable through a structured JSON audit trail. And an interactive demonstration was produced, featuring a disease progression slider, a live clinical reasoning panel, and per-case explanations that update in real time.

The system correctly differentiates all eight cases across the severity spectrum from Case 08 with a progression score of 0.218 and near-healthy visualisation, to Case 01 with a progression score of 0.809 and advanced disease morphology showing substantial volume loss, dark fibrotic colouration, and pronounced structural deformation. The audit logs confirm deterministic, reproducible behaviour across all cases, with transformation parameters scaling monotonically with disease severity.

6.4 Significance of Results

The significance of these results lies at two levels. At the technical level, the project demonstrates that a small set of interpretable CT features computable using standard image processing techniques without any deep learning infrastructure is sufficient to drive a 3D disease visualisation that produces clearly differentiated, stage-appropriate, and clinically grounded outputs across a real patient cohort. This is a non-trivial finding: it suggests that the complexity of medical visualisation systems does not need to scale with the complexity of their inputs, and that transparent, rule-based mapping can produce results competitive with more opaque generative approaches for educational and communication purposes.

At the broader level, the project demonstrates a working architecture for explainable medical visualisation, one in which every output is auditable, every transformation is traceable, and every decision can be explained in plain language. This architecture is directly relevant to the growing conversation in healthcare AI about what clinical accountability means in practice, and this project provides a concrete prototype of what it can look like in a real system.

6.5 Future Work

Several significant directions for future development were identified during this project.

The most impactful near-term extension would be the integration of full volumetric CT processing. The current system extracts features from a single 2D cross-sectional slice per case. Processing multiple slices across the full lung volume and aggregating signals three-dimensionally would substantially improve the clinical credibility of the extracted features and would allow the system to represent spatially specific disease patterns such as upper versus lower lobe involvement rather than a single aggregate severity measure.

The second priority would be the integration of Chirag Srinivasamurthy's ClinicalBERT text extraction component as an optional input mode. The structured JSON output format he developed specifying density, region, scarring, pattern, severity, and deformation intensity maps directly onto the visualisation parameters already present in the pipeline. Implementing a translation layer between this format and the existing parameter space would add text-driven input capability without requiring changes to the core simulation or rendering code.

A third important direction is clinical validation. The system currently has no mechanism for assessing how well its stage assignments and visual outputs correspond to ground truth clinical assessments. Establishing a validation framework working with radiologists or pulmonologists to annotate the eight cases with expected stages, dominant disease regions, and approximate extent of fibrotic involvement would provide the quantitative evidence needed to evaluate the system's clinical utility and to identify specific areas where the mapping model requires refinement.

A fourth direction is the development of patient-specific geometry. The current system applies all deformations to a single reference mesh from the Human Atlas API. Incorporating patient-specific lung geometry derived from the CT data itself using segmentation-based

surface reconstruction would allow the system to represent anatomical variation between patients and to localise disease effects to the specific regions where the CT imaging shows abnormality, rather than distributing them according to a generic anatomical weighting scheme.

Finally, the extension of the system to other organs and other diseases building on the same pipeline architecture would demonstrate the generalisability of the approach and increase its value as a platform for medical education and communication. The Human Atlas API provides reference meshes for a range of organs, and the signal extraction, normalisation, and mapping framework developed in this project is not specific to pulmonary fibrosis.

Bibliography

- Bhadra, S., Kelkar, V.A., Brooks, F.J. & Anastasio, M.A. 2021, 'On Hallucinations in Tomographic Image Reconstruction', *IEEE Transactions on Medical Imaging*, vol. 40, no. 11, pp. 3249-3260.
- Devlin, J., Chang, M.-W., Lee, K. & Toutanova, K. 2019, 'BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding', *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (NAACL-HLT)*, pp. 4171-4186.
- Dinc, M.T., et al. 2025, 'Large Language Models in Healthcare and Medical Applications: A Review', *PubMed Central*, vol. 12, article PMC12189880.
- Huang, K., Altosaar, J. & Ranganath, R. 2019, 'ClinicalBERT: Modeling Clinical Notes and Predicting Hospital Readmission', *arXiv preprint arXiv:1904.05342*.
- Huang, L., Yu, W., Ma, W., Zhong, W., Feng, Z., Wang, H., Chen, Q., Peng, W., Feng, X., Qin, B. & Liu, T. 2025, 'A Survey on Hallucination in Large Language Models: Principles, Taxonomy, Challenges, and Open Questions', *ACM Computing Surveys*, vol. 57, no. 2, pp. 1-35.
- Weatherley, N. D., Eaden, J. A., Stewart, N. J., Bartholmai, B. J., Swift, A. J., Bianchi, S. M., & Wild, J. M. (2019). Experimental and quantitative imaging techniques in interstitial lung disease. *Thorax*, 74(6), 611–619.
- Jones, M.G., Fabre, A., Schneider, P., Cinetto, F., Sgalla, G., Mavrogordato, M. & Davies, D.E. 2016, 'Three-dimensional Characterisation of Fibroblast Foci in Idiopathic Pulmonary Fibrosis', *JCI Insight*, vol. 1, no. 5, article e86375.
- Lu, H. & Naseem, U. 2024, 'Can Large Language Models Enhance Predictions of Disease Progression? Investigating Through Disease Network Link Prediction', *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pp. 1120-1135.
- Lybarger, K., Damani, A., Gunn, M.L., Uzuner, O. & Yetisgen, M. 2021, 'Extracting Radiological Findings with Normalised Anatomical Information Using a Span-based BERT Relation Extraction Model', *Journal of Biomedical Informatics*, vol. 117, article 103740.
- Sun, X., Jia, S., Pan, Z., Wu, K. & Bera, A. 2025, 'Handle-based Mesh Deformation Guided by Vision Language Model', *ResearchGate*, preprint.
- Tavakoli, N. & Kim, D. 2025, 'AI-generated Clinical Histories for Radiology Reports: Closing the Information Gap', *Radiology*, vol. 314, no. 2, article e243910.
- Trinh, Q.-H., Ding, X., Liu, Y., Qin, Z., Li, X., Durak, G., Aktas, H.E., Keles, E., Bagci, U. & Xu, M. 2026, 'Beyond Medical Diagnostics: How Medical Multimodal Large Language Models Think in Space', *BioRxiv preprint*.

Tustison, N.J., Avants, B.B. & Stone, J.R. 2024, 'Biomedical Large Language Model Latent Spaces for Representing Disease Phenotypes and Pseudotime', *MedRxiv preprint*.

Bompoti, A., Papazoglou, A. S., Moysidis, D. V., Otountzidis, N., Karagiannidis, E., Stalikas, N., Panteris, E., Ganesh, V., Sanctuary, T., Arvanitidis, C., Sianos, G., Michaelson, J. S., & Herrmann, M. D. (2021). Volumetric Imaging of Lung Tissue at Micrometer Resolution: Clinical Applications of Micro-CT for the Diagnosis of Pulmonary Diseases. *Diagnostics*, 11, 2075. <https://www.mdpi.com/2075-4418/11/11/2075>

Wang, A. Q., Karaman, B. K., Kim, H., Rosenthal, J., Saluja, R., Young, S. I., & Sabuncu, M. R. (2024). A Framework for Interpretability in Machine Learning for Medical Imaging. *IEEE Access*, 12, 53277-53292.

Appendix

Github Repository

Code to the visualisation prototype, data ingestion pipelines as well as the Blender application can be found on GitHub via the following link:

<https://github.com/MSkandula/Disease-Progression-visuliazation.git>

GitHub Repository Link

The complete source code, methodology implementation, audit logs, and supporting project files are available at:

https://github.com/anil-0618/final_methodology